CONSUMER RISING

The Label Decoder

53 ingredients to watch — and the exact words to look for on the label, in plain language.

“Know what goes into your body.”

A Consumer Rising field guide · consumerrising.com · Free forever

How to use this guide

You don’t need to memorize 53 ingredients. You need to recognize the handful that matter most — right there in the grocery aisle.

1 **Start with Avoid**
These 18 are the ones with the most serious questions around them. Learn these first; the rest can wait.

2 **Use the “Look for” line**
Every entry lists the exact label words — aliases, E-numbers, trade names — so you can match what you see on the package.

3 **Scan, don’t memorize**
Pair this guide with the free Consumer Rising label scanner: point your phone at any ingredients panel and it flags these automatically.

THE FIVE LEVELS

AVOID The ingredients with the most serious questions around them — the ones worth learning first.

CAUTION Strong reasons to be careful. Worth limiting or skipping when you have a choice.

CERTAIN PEOPLE SHOULD AVOID Fine for most people — but specific groups (kids, pregnant women, people with certain conditions) should steer clear.

CUT BACK Not poisons — dose problems. Fine in small amounts; the trouble is how much of them modern food contains.

LOWER CONCERN Lower on the worry list — included so you can recognize the names and move on.

These levels are Consumer Rising’s editorial judgment, not medical advice. Every entry links to a full dossier with primary sources at consumerrising.com/ingredients.

AVOID

The ingredients with the most serious questions around them — the ones worth learning first.

Acesulfame Potassium (Ace-K)

AVOID

SWEETENER · E950

Look for: Acesulfame Potassium, Acesulfame-K, Ace-K, Sunett, E950

A calorie-free sweetener found in diet sodas, sugar-free baked goods, and other 'diet' products. It is FDA-approved, but CSPI rates it Avoid — and a large 2022 French cohort linked higher intake to higher coronary heart disease risk, a signal worth knowing about but not proof of harm.

Aspartame

AVOID

SWEETENER · E951

Look for: E951, NutraSweet, Equal

The sweetener in most diet sodas and sugar-free gum. In 2023 the WHO's cancer agency called it 'possibly carcinogenic' — while other expert groups disagreed.

Azodicarbonamide (ADA)

AVOID

DOUGH CONDITIONER · E927A

Look for: azodicarbonamide, ADA, azobisformamide (related), E927a, E927

The 'yoga mat chemical' — a dough conditioner that makes bread rise higher and fluffier, best known from the 2014 campaign that got Subway to drop it. When baked, it breaks down into semicarbazide.

BHA

AVOID

ANTIOXIDANT · E320

Look for: Butylated hydroxyanisole, E320

A preservative that keeps fats in chips, cereal, and gum from going stale. Listed as 'reasonably anticipated to be a human carcinogen' in the US.

Blue 1 (Brilliant Blue)

AVOID

ARTIFICIAL DYE · E133

Look for: Blue 1, Brilliant Blue, Brilliant Blue FCF, FD&C Blue No. 1, E133

The blue in candy, sports drinks, and baked goods — often mixed with yellows to make green. It carries the same class-level concerns as the other synthetic dyes, but the honest truth is that Blue 1 has thinner dye-specific research than the yellows and reds.

Brominated Vegetable Oil (BVO)

AVOID

EMULSIFIER · E443

Look for: BVO, E443

Used to keep citrus flavor mixed in some sodas and sports drinks. The FDA revoked its authorization for use in food in 2024.

Partially Hydrogenated Oils

AVOID

FAT

Look for: PHOs, Partially hydrogenated soybean oil, Shortening (some)

The main source of artificial trans fat. The FDA gave the industry until 2020 to remove them — but label loopholes mean traces can remain.

Potassium Bromate

AVOID

DOUGH CONDITIONER · E924

Look for: Bromated flour, E924

A flour additive that makes bread rise higher and look whiter. Removed from EU food years ago — still allowed in the US.

Propyl Gallate

AVOID

ANTIOXIDANT · E310

Look for: propyl gallate, E310

An antioxidant that protects fats and oils in meat products, frozen foods, and candy — usually teamed up with BHA and BHT. EWG calls it an ingredient of concern and recommends limiting it.

Red 3 (Erythrosine)

AVOID

ARTIFICIAL DYE · E127

Look for: Red 3, Erythrosine, FD&C Red No. 3, Red No. 3, E127

The bright cherry-red dye in candy, frostings, and baked goods — and the additive behind the biggest US food-dye regulatory action in decades. In January 2025 the FDA revoked its authorization for food, but be clear-eyed about why: the decision was a matter of law, not new evidence of human harm.

Red 40

AVOID

ARTIFICIAL DYE · E129

Look for: Allura Red AC, E129, FD&C Red No. 40

One of the most widely used artificial colors in US food — candy, drinks, cereals, and snacks. It makes food look bright red or orange.

Saccharin

AVOID

SWEETENER · E954

Look for: Saccharin, Sweet'N Low, sodium saccharin, E954

A zero-calorie artificial sweetener with a long, instructive history: a 1970s rat bladder-cancer scare put warning labels on every packet, then the mechanism turned out to be rat-only and the labels came off in 2000.

Sodium Nitrite & Sodium Nitrate

AVOID

PRESERVATIVE · E250

Look for: Sodium Nitrite, Sodium Nitrate, E250, E251, E249 (Potassium Nitrite), E252 (Potassium Nitrate), celery powder, celery juice powder, cultured celery juice

The preservatives that keep bacon pink, hot dogs shelf-stable, and deli meat free of botulism. They do a real food-safety job — but in the body's acid stomach environment, nitrite can convert into nitrosamines, and IARC classifies ingested nitrite under nitrosating conditions as probably carcinogenic to humans (Group 2A).

Sucralose

AVOID

SWEETENER · E955

Look for: Sucralose, Splenda, E955

The sweetener sold as Splenda, spread across the 'sugar-free' aisle — diet drinks, gum, baked goods, and tabletop packets. Two independent 2022 studies raised flags: a large French cohort linked it to coronary heart disease risk, and a human trial found it disrupted gut bacteria and blood-sugar response.

TBHQ (Tertiary Butylhydroquinone)

AVOID

ANTIOXIDANT · E319

Look for: TBHQ, tertiary butylhydroquinone, tert-butylhydroquinone, t-butylhydroquinone, E319

The preservative that keeps crackers, potato products, and oil-heavy baked goods from going rancid — EWG found it in over 2,700 packaged foods. Human data on its effects is essentially absent, which is itself the point.

Titanium Dioxide

AVOID

COLOR · E171

Look for: E171, CI 77891

Makes candy, frosting, and coffee creamer look bright white. The EU banned it as a food additive in 2022 after EFSA's safety review.

Yellow 5

AVOID

ARTIFICIAL DYE · E102

Look for: Tartrazine, E102, FD&C Yellow No. 5

The bright yellow dye in candy, drinks, chips, and even some medicines. It carries the same EU child-attention warning as Red 40.

Yellow 6 (Sunset Yellow)

AVOID

ARTIFICIAL DYE · E110

Look for: Yellow 6, Sunset Yellow, Sunset Yellow FCF, FD&C Yellow No. 6, E110

One of the most widely used dyes in US candy, sodas, and snacks. California's 2021 OEHHA review — the most rigorous assessment of dyes and children to date — found synthetic dyes can worsen neurobehavioral problems in some children, and that the FDA's intake limits may not protect kids.

CAUTION

Strong reasons to be careful. Worth limiting or skipping when you have a choice.

BHT (Butylated Hydroxytoluene)

CAUTION

ANTIOXIDANT · E321

Look for: BHT, butylated hydroxytoluene, butylhydroxytoluene, E321

A synthetic antioxidant that keeps fats and oils in cereals, snack foods, and packaged goods from going rancid. It holds a spot on EWG's Dirty Dozen list of additives to avoid — though the case against it is precautionary, not proven.

Blue 2

CAUTION

ARTIFICIAL DYE · E132

Look for: Blue 2, FD&C Blue No. 2, indigotine, E132

One of the less-studied synthetic blue dyes, used in candy, drinks, baked goods, and pet food. It carries the class-level concerns behind all synthetic dyes without much dye-specific data of its own.

Caramel Color III & IV

CAUTION

COLOR · E150C

Look for: caramel color, caramel colour, caramel coloring, caramel colouring, ammonia caramel (related), sulfite ammonia caramel (related), E150c, E150d

The brown in colas, soy sauce, and baked goods — the most widely used coloring in food by weight. Only the ammonia-processed kinds (Classes III and IV) carry the contaminant 4-MEI.

Carboxymethylcellulose

CAUTION

THICKENER · E466

Look for: Carboxymethylcellulose, cellulose gum, sodium carboxymethylcellulose, E466

A thickener and emulsifier that gives ice cream its body and keeps baked goods and sauces smooth — backed by two independent lines of cautionary evidence.

Carrageenan

CAUTION

THICKENER · E407

Look for: E407, Irish moss (source)

A thickener from red seaweed in chocolate milk, ice cream, and plant milks. Lab research raised gut-inflammation questions; regulators say current uses look safe.

Diacetyl

CAUTION

FLAVORING

Look for: diacetyl, 2,3-butanedione, artificial butter flavor, artificial butter flavoring

The chemical behind the butter flavor in microwave popcorn and many baked goods — the one behind 'popcorn lung.'

Interesterified Oils

CAUTION

FAT

Look for: interesterified soybean oil, interesterified palm oil, interesterified oils

The industry's quiet replacement for trans fats in margarines, baked goods, and confectionery — made by chemically rearranging which fatty acids sit on the fat molecule.

Mono- and Diglycerides of Fatty Acids

CAUTION

EMULSIFIER · E471

Look for: Mono- and diglycerides, monoglycerides, distilled monoglycerides, mono- & diglycerides of edible fats or oils, E471

One of the most widely consumed emulsifiers in processed food — it keeps cakes, biscuits, mayonnaise, and other fats and sauces smooth and shelf-stable.

Parabens (Propylparaben & Methylparaben)

CAUTION

PRESERVATIVE · E216

Look for: propylparaben, propyl paraben, methylparaben, methyl paraben, propyl p-hydroxybenzoate (related), E216, E218

Antimicrobial preservatives in tortillas, muffins, and some processed meats — the same chemical family used in cosmetics. Europe banned propylparaben from food in 2006; the US never did.

Polysorbate 80

CAUTION

EMULSIFIER · E433

Look for: Polysorbate 80, polysorbate-80, tween 80, polyoxyethylene (20) sorbitan monooleate, E433

An emulsifier used in ice cream, baked goods, and creamy sauces — and one of two emulsifiers tested in a much-discussed 2015 mouse study on gut inflammation.

Sodium Aluminum Phosphate

CAUTION

LEAVENING ACID · E541

Look for: sodium aluminum phosphate, SALP, E541

The leavening acid in self-rising flour, baking powder, cake mixes, and processed cheese. It is one of the main food-additive sources of dietary aluminum.

CERTAIN PEOPLE SHOULD AVOID

Fine for most people — but specific groups (kids, pregnant women, people with certain conditions) should steer clear.

Artificial Flavors

CERTAIN PEOPLE SHOULD AVOID

FLAVORING

Look for: Artificial flavors, Artificial flavor, Artificial flavoring

A catch-all label word for flavor chemicals made in a lab rather than pulled from food — and you usually cannot tell what is actually inside it.

Carmine

CERTAIN PEOPLE SHOULD AVOID

COLOR · E120

Look for: Carmine, Cochineal extract, Cochineal, Carminic acid, Natural Red 4, E120

A pink-to-red color made from the cochineal insect — the 'natural' color with a documented, if rare, severe-allergy risk.

Erythritol

CERTAIN PEOPLE SHOULD AVOID

SWEETENER · E968

Look for: Erythritol, E968

A sugar alcohol marketed in keto and 'sugar-free' products as the safe sweetener — now at the center of a live scientific debate about heart risk.

Monosodium Glutamate (MSG)

CERTAIN PEOPLE SHOULD AVOID

FLAVOR ENHANCER · E621

Look for: MSG, E621, Yeast extract (similar), Hydrolyzed protein (similar)

A flavor booster for savory food. The scare started in 1968; the science since then has been mixed and mostly reassuring at normal food levels.

Natural Flavors

CERTAIN PEOPLE SHOULD AVOID

FLAVORING

Look for: natural flavors, natural flavoring, natural flavouring

The most common two-word mystery on US labels — a legal category, not a health claim, covering dozens of undisclosed flavor chemicals derived from natural sources.

Phosphates (incl. Phosphoric Acid)

CERTAIN PEOPLE SHOULD AVOID

EMULSIFIER · E338

Look for: phosphoric acid, sodium phosphate, sodium acid pyrophosphate, trisodium phosphate, dicalcium phosphate, E338, E339, E450, E451

A huge family of additives doing dozens of jobs — leavening baked goods, emulsifying processed cheese, adding tang to colas. One recent study found phosphate additives in 56% of products from America's top 25 food makers.

Propylene Glycol

CERTAIN PEOPLE SHOULD AVOID

SOLVENT · E1520

Look for: Propylene glycol, Propane-1,2-diol, E1520

A moisture-holding solvent that keeps cake mixes soft and carries flavors and colors evenly through food — a different chemical from the toxic antifreeze ingredient it keeps getting confused with.

Sodium Benzoate

CERTAIN PEOPLE SHOULD AVOID

PRESERVATIVE · E211

Look for: E211, Benzoic acid (related)

A preservative that keeps sodas, juices, and condiments from spoiling. Common in acidic drinks.

Sulfites

CERTAIN PEOPLE SHOULD AVOID

PRESERVATIVE · E220

Look for: Sulfites, sulfur dioxide, sodium sulfite, sodium bisulfite, sodium metabisulfite, potassium bisulfite, potassium metabisulfite, E220, E223

Preservatives that keep dried fruit bright and wine from spoiling — safe for most people, but genuinely dangerous for asthmatics and sensitive individuals.

Xylitol

CERTAIN PEOPLE SHOULD AVOID

SWEETENER · E967

Look for: Xylitol, birch sugar, E967

A sugar alcohol often derived from birch, used in sugar-free candy, 'keto-friendly' baked goods, and toothpaste — the second sugar alcohol flagged by Cleveland Clinic researchers for clotting risk.

CUT BACK

Not poisons — dose problems. Fine in small amounts; the trouble is how much of them modern food contains.

Allulose

CUT BACK

SWEETENER

Look for: allulose, D-allulose, D-psicose, psicose

The new 'rare sugar' sweetener in keto products — about a tenth of sugar's calories, without the same blood-sugar spike.

Caffeine

CUT BACK

STIMULANT

Look for: 1,3,7-Trimethylxanthine

The world's most-used stimulant — in coffee, tea, cola, energy drinks, and chocolate. Fine in moderation for most adults; easy to overdo without noticing.

High Fructose Corn Syrup

CUT BACK

SWEETENER

Look for: HFCS, HFCS-55, Glucose-fructose syrup, Isoglucose

A cheap liquid sweetener made from corn starch. It is the main sweetener in most American sodas and shows up in bread, ketchup, and salad dressing too.

Salt (Sodium Chloride)

CUT BACK

SEASONING

Look for: Sodium chloride, Table salt

The world's oldest seasoning and preservative — essential in small amounts, but most people get far more than they need, and most of it comes from packaged foods, not the salt shaker.

Sorbitol

CUT BACK

SWEETENER · E420

Look for: Sorbitol, E420, Sorbitol syrup

A sugar alcohol used to sweeten sugar-free gum, candy, and diet foods — the concern here is concrete and well documented: too much upsets your stomach.

Sugar (Sucrose)

CUT BACK

SWEETENER

Look for: Sucrose, Table sugar, Cane sugar

Table sugar — half glucose, half fructose. The body treats it much like high fructose corn syrup; the problem is the total amount, not the source.

LOWER CONCERN

Lower on the worry list — included so you can recognize the names and move on.

Calcium Disodium EDTA

LOWER CONCERN

PRESERVATIVE · E385

Look for: Calcium disodium EDTA, Calcium disodium edetate, EDTA, E385

A preservative that works by grabbing onto stray metal ions — tiny amounts of iron and copper in food that would otherwise turn fats rancid and colors dull.

Calcium Propionate

LOWER CONCERN

PRESERVATIVE · E282

Look for: calcium propionate, E282, preservative 282

The mold inhibitor behind long-lasting bread — the quiet preservative in the bread aisle.

DATEM

LOWER CONCERN

DOUGH CONDITIONER · E472E

Look for: DATEM, E472e, diacetyl tartaric acid esters of mono- and diglycerides

One of the most common dough conditioners in commercial bread and buns — it makes loaves rise higher, stay soft longer, and survive high-speed industrial baking.

Dimethylpolysiloxane

LOWER CONCERN

PROCESSING AID · E900

Look for: dimethylpolysiloxane, polydimethylsiloxane, PDMS, E900, silicone oil

A silicone-based anti-foaming agent that keeps hot cooking oil from foaming over — a processing aid you will likely never taste or notice.

Disodium Inosinate & Disodium Guanylate

LOWER CONCERN

FLAVOR ENHANCER · E631

Look for: disodium inosinate, disodium guanylate, E631, E627, I+G

The umami amplifiers hiding behind MSG on snack and soup labels — they make savory flavors taste several times stronger than MSG alone.

Green 3

LOWER CONCERN

ARTIFICIAL DYE · E143

Look for: Green 3, Fast Green FCF, FD&C Green No. 3, E143

The quiet one of the synthetic dye family — a green color used in some candies and beverages, and not widely used at all.

Maltodextrin

LOWER CONCERN

THICKENER

Look for: Maltodextrin

A highly processed starch used as a filler, thickener, and carrier in everything from salad dressing to 'sugar-free' snacks.

Sodium Stearoyl Lactylate

LOWER CONCERN

EMULSIFIER · E481

Look for: sodium stearoyl lactylate, SSL, E481, sodium stearoyl-2-lactylate

DATEM's quieter partner in the bread aisle — an emulsifier that strengthens dough, softens crumb, and keeps commercial loaves fresh longer.

What now?

You now know the 53 names that matter most on American labels — and the exact words they hide behind. Keep this guide on your phone, or print it and keep it where you unpack groceries.

For the full story behind any ingredient — the evidence, the regulatory history, the primary sources — visit the Ingredient Index at **consumerrising.com/ingredients**. Every dossier is free, cited, and updated as the science moves.

“Know what goes into your body.”